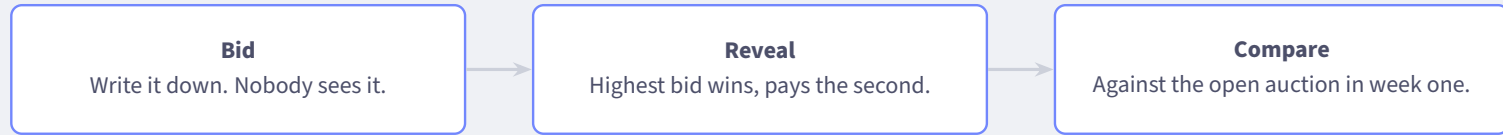


Auctions

The same five pounds, sold a different way



On sale: a five pound note. What is it worth to you?

Bidder	Bid

Bidder	Bid

Winner

Highest bid

Price paid

Their profit

Everyone submits _____, without seeing anyone else's.

The item goes to _____.

The winner pays _____.

In week one we ran this open and ascending, and it went badly. Name one thing that changed when we sealed it:

Your value is v . Let b be the highest bid among everybody else. Compare bidding v with bidding something else.

You bid above your value

You only win extra times when b is between and
, and then you pay and make .

So bidding your true value is _____.

You bid below your value

You lose some auctions you would have won, and when you do still win you pay .

If truth is a weakly dominant strategy, then to bid well you need to know _____ about the other bidders.

So the seller gets the item to the person who _____, which is not guaranteed otherwise.

Now change one rule: the winner pays their *own* bid. Is truth still weakly dominant?

What should you do instead?

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What we just did in the room is **Question 1** on the Auctions page, with values $v_1 = 10$, $v_2 = 7$ and $v_3 = 4$.

- (a) The rules of a second-price (Vickrey) auction. [3]
- (b) With truthful bids: the winner, the price and the payoff. [3]
- (c) Show bidding $v_1 = 10$ is weakly dominant, above and below. [7]
- (d) Why that is a desirable property. [4]
- (e) Would truth still be weakly dominant in a first-price auction? [2]

vknights.org/gt/topics/auctions.html